Iteration #1

1. Wearable heart rate monitor
2. Medical device simulator
3. Electrical vehicle charging station

Objectives

- Loop and “count” the times of when a “heart-beat” is detected, also keep track of time passed
- Calculate BPM from the beats per minute
- Keep a specific amount of recent readings -> use readings to find patterns for conditions/diseases

Vivian:

1. Sensor, wiring, power, enclosure
2. Drawing the circuit diagram, wiring the sensor and board, writing code to read raw sensor values, designing the power supply and battery.
3. Deliverable: working hardware that outputs clean raw readings.

Kai:

1. Signal processing, BPM logic, display and alerts
2. Install libraries in Arduino IDE, write beat detection and BPM algorithm (threshold and moving average).
3. Program the OLED display and alert logic.
4. Deliverable: Firmwave that

Shared Tasks

1. Integration testing, accuracy validation, documentation, final presentation.
2. Planning
3. Researching thresholds, project design, circuit schematic/design

Technologies & Tools

Tools and Rational:

1. Arduino IDE - uses simplified c++ with ready-made functions such as analogRead(). A working heart rate reading can come out of a few lines. There is a PulseSenor Playground for beast detection, wire hardware support and a built in serial monitor.
2.

Category	**Choice**
Development Environment	Arduino IDE, direct control over pins, timing, sensor.
Microcontroller Board	Arduino Nano, small, cheap, low power which suits a wearable.
Sensor	PulseSensor (analog), inexpensive, well documented, and has mature libraries.
Display	0.96 “ SSD1306 OLD (I2C), low power and readable so it fits on a wearable without needing a mobile device.
Libraries	Arduino core framework, Wire (I2C), SparkFun MAX3010x, Adafruit SSD1306 and Adafruit GFXD, avoids writing low-level I2C drivers.
Debugging tools	Serial Monitor and Serial Plotter
Version control and collaboration	GitHub, version control prevents overwritten work and keeps a history of changes.
Circuit Design	LTSpice, learning LTSpice in biomedical applications.
Testing and validation	Comparison with applewatch, logged into excel, manual measurement tests

Timeline:

Vivian

Kai

Week	Task	Deliverable
1	Iteration 1 Research sensors, boards, select components, set up GitHub Repo	Written objectives and scope
2	Design circuit and define pin map Get components and test each Write sensor reading sketch Write OLED display test and basic UI layout	Circuit Diagram on LTSpice Videos of each component working Sketch that displays sample BPM on OLED
3	Assemble prototype on breadboxes Develop beat detection and BPM calculation (threshold/peak detection) Add moving average and unrealistic value rejection	Working breadboard prototype BPM algorithm sketch tested with a sample or live data Smoothed, stable BPM output
4	Both: Integrate hardware and software and debug noise, finger-placement, and timing issues. Add alert logic	Both: First working prototype showing live BPM and Bug and fix log. Alert feature working and demonstrated. Powered prototype running on battery.

	(buzzer/LED for BPM out of range) B Design power circuit (batter, charger, switch)	
5	Build compact version (breadboard or soldered) Optimise code (memory, power use, display refresh)	Soldered, compact circuit Clean, commented final firmware
6	Both: Plan and run accuracy tests against a reference device and fix issues found in testing, retest Calculate error percentage and analyse results Test battery life and wearability	Test results spreadsheet and improved version and retest results Accuracy analysis with charts Battery and durability test notes
7	Complete final report and prepare presentation and rehearse demo Final code documentation (Kai) and wiring documentation (Vivian) Final demonstration	Final project report, slides and demo script README and commented code (Kai), circuit diagram (Vivian) Live demo of working wearable monitor

Milestones:

Week 1: Objectives and component list approved
Week 2: Circuit designed and sensor reading raw data
Week 3: Breadboard prototype and BPM algorithm working
Week 4: Integrated prototype with alerts and battery power
Week 5: Compact wearable built
Week 6: Accuracy testing complete
Week 7: Report submitted and demo delivered

Communication:

1. Group Chat and Github.
2. Weekly meets for progress checks.
3. We will create meeting notes to address and implement changes to our program.

Contributions for this iteration:

Vivian	Kai
Tools and explanation, contributions, milestones and timeline	Objectives, researching a wearable heart rate monitor, suggestions for coding environment